

Operations on functions

Building new functions out of old ones, reading a graph, and the rules of logarithms.

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- 2.1 Adding and multiplying functions
 - 2.2 Reading a graph
 - 2.3 Composition of functions
 - 2.4 Inverses, and when one exists
 - 2.5 Finding an inverse
 - 2.6 The rules of logarithms
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2.1 Adding and multiplying functions

LEARN IT ON KHAN ACADEMY

Khan has no lesson on $f + g$, $f - g$ and fg as such. These two are the same algebra on expressions, which is the closest it gets.

PRACTISE IT ON KHAN ACADEMY

 **Exercise** Quotients and sums of fractions · Add & subtract rational expressions

 **Exercise** Products of functions · Multiply binomials

COMBINING FUNCTIONS

Two functions can be added, subtracted, multiplied or divided at every input they share.

$$(f + g)(x) = f(x) + g(x) \quad (fg)(x) = f(x)g(x)$$

and the same for the difference, the quotient, and a constant times a function.

WHERE IT IS DEFINED

Only where both originals are. For a quotient, also nowhere that $g(x) = 0$.

- 1 **Work out each function separately at the input.** Two numbers, not one expression.

- 2 **Combine those two values with the operation.**
- 3 **Intersect the domains.** The combination exists only where both originals do.
- 4 **For a quotient, exclude anything that makes the bottom zero.** Even where the top is perfectly well behaved.
- 5 **Bracket the second function before you subtract it.** $-(x^2 + 3x) = -x^2 - 3x$. Both signs change.

$$F(x) = 3x - 1 \text{ AND } G(x) = x^2 + 3x$$

$$(f - g)(x) = 3x - 1 - (x^2 + 3x)$$

Bracket g before subtracting.

$$= -x^2 - 1$$

The 3x cancels.

$$\left(\frac{f}{g}\right)(x) = \frac{3x - 1}{x^2 + 3x}$$

No cancelling: nothing is a common factor.

undefined at $x = 0$ and $x = -3$

Because $x^2 + 3x = x(x + 3)$ is zero there.

THE ONE YOU WILL USE ALL YEAR

$$P(q) = R(q) - C(q)$$

Profit is revenue minus cost, which is a subtraction of two functions and nothing more. Cost itself is usually a sum: a fixed part plus a variable part.

WATCH OUT

- Subtracting without brackets. It changes the sign of every term after the first.
- Cancelling across a plus. $\frac{3x - 1}{x^2 + 3x}$ does not simplify.
- Forgetting that the combination inherits both domains, not just the more generous one.

2.2 Reading a graph

LEARN IT ON KHAN ACADEMY

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- ▶ **Video** Both intercepts · x-intercepts and y-intercepts
 - ▶ **Video** Intercepts of a parabola · Interpret a quadratic graph
 - ≡ **Article** Intercepts of lines review (x-intercepts and y-intercepts)
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- ✍ **Exercise** From the formula · Intercepts from an equation
 - ✍ **Exercise** Intercepts from a graph
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INTERCEPTS

The y-intercept is the value at zero, $f(0)$. There is at most one, because a function gives one output per input. An x-intercept, or a zero, is any value where $f(x) = 0$. There may be several, or none.

- 1 **For the y-intercept, put $x = 0$.** One substitution, no solving.
- 2 **For the x-intercepts, solve $f(x) = 0$.** This is the step that is actual work.
- 3 **Plot the intercepts first.** They are the two easiest points on any curve.
- 4 **Add enough further points to see the shape.** For a line, two points are the whole story.
- 5 **Read the intercepts back into the question.** A y-intercept is usually a fixed cost or a starting value. An x-intercept is usually a break-even point or the moment something runs out.

$$Y = 2X + 3$$

$$\text{y-intercept: } y = 2(0) + 3 = 3$$

The curve crosses the vertical axis at 3.

$$\text{x-intercept: } 2x + 3 = 0$$

Set the output to zero and solve.

$$x = -\frac{3}{2}$$

One crossing, because a line has at most one.

A VERTICAL LINE IS NOT A FUNCTION

$x = 3$ is a perfectly good line and has no y-intercept to speak of. It fails the vertical line test, so it is not the graph of a function of x .

WATCH OUT

- Solving $f(x) = 0$ and reporting the output. The answer is the input.
- Assuming there is one x-intercept. A parabola has two, one, or none.
- Reading an intercept off a sketch instead of solving for it.

2.3 Composition of functions

LEARN IT ON KHAN ACADEMY

- ▶ **Video** What a composite is · Composing functions

- ▶ **Video** At a number · Evaluating composite functions

- ≡ **Article** The definition · Intro to composing functions

PRACTISE IT ON KHAN ACADEMY

- ✍ **Exercise** At a number · Evaluate composite functions

 - ✍ **Exercise** Building the formula · Find composite functions
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COMPOSITION

$$(f \circ g)(x) = f(g(x))$$

Do g first, then put its output into f . The circle is read right to left, which is the single most common thing to get wrong.

WHERE IT IS DEFINED

x has to be something g accepts, **and** $g(x)$ has to be something f accepts. Both conditions, not either.

- 1 **Evaluate the inner function at the input.** One number out.

- 2 **Put that output into the outer function.** That is the whole procedure for a number.

- 3 **For a formula, substitute the entire inner expression.** Every x in the outer function becomes the whole of $g(x)$, brackets included.

- 4 **Check the order.** $f \circ g$ is almost never $g \circ f$. One number settles it.

- 5 **For the domain, take what g accepts, then remove anything whose output f refuses.**

$$F(X) = 2X + 1 \text{ AND } G(X) = X^2 - 3$$

$$(g \circ f)(4): f(4) = 9$$

Inner function first.

$$g(9) = 81 - 3 = 78$$

Its output is the input to the outer one.

$$(f \circ g)(x) = 2(x^2 - 3) + 1$$

Symbolically now: the whole of g goes in, in brackets.

$$= 2x^2 - 5$$

Expand afterwards.

WHY IT TURNS UP

Advertising spend gives visitors, visitors give revenue. The question asks revenue from advertising spend. The number of visitors is real, and the question is not about it.

WATCH OUT

- Reading $f \circ g$ left to right. The inner function is the one on the right.
- Substituting without brackets, so only part of the inner expression gets squared or multiplied.
- Assuming the domain is just what g accepts. The middle value has to be legal too.

2.4 Inverses, and when one exists

LEARN IT ON KHAN ACADEMY

- ▶ **Video** What an inverse is · Intro to inverse functions

- ▶ **Video** When one exists · Invertible functions

- ≡ **Article** One-to-one, written out · Intro to invertible functions

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- ✍ **Exercise** Spotting when there is no inverse · Determine if a function is invertible

 - ✍ **Exercise** Reading an inverse off a graph · Inverse functions: graphs and tables
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INVERSE FUNCTION

For a one-to-one function f , its inverse f^{-1} sends every output back to the input it came from.

$$f^{-1}(f(x)) = x \quad f(f^{-1}(x)) = x$$

ONE-TO-ONE

Different inputs always give different outputs. No output is ever produced twice. This is the whole condition for an inverse to exist.

DOMAIN AND RANGE SWAP

The range of f is the domain of f^{-1} , and the domain of f is the range of f^{-1} .

- 1 **To test two functions, compose them both ways.**
- 2 **Work out $f(g(x))$ and $g(f(x))$.** Both, not one.
- 3 **If both come back to x , they are inverses. If either does not, they are not.**
- 4 **To decide whether an inverse exists at all, use the horizontal line test.** A horizontal line may cross the graph at most once. Twice means two inputs share an output, so there is no way back.
- 5 **A function that only ever rises, or only ever falls, passes.** Strictly increasing or strictly decreasing is enough to be one-to-one.

6 The graph of the inverse is the reflection in the line $y = x$.

ARE $F(x) = 2x - 1$ AND $G(x) = \frac{x + 1}{2}$ INVERSES?

$$f(g(x)) = 2\left(\frac{x + 1}{2}\right) - 1$$

Compose one way.

$$= x + 1 - 1 = x$$

Back to x .

$$g(f(x)) = \frac{(2x - 1) + 1}{2}$$

Now the other way.

$$= x$$

Both directions return x , so they are inverses.

WATCH OUT

- Reading f^{-1} as a reciprocal. It is not $\frac{1}{f(x)}$.
- Composing one way only. One direction can work by accident.
- Assuming every function has an inverse. $f(x) = x^2$ does not, until you restrict it.

2.5 Finding an inverse

LEARN IT ON KHAN ACADEMY

► **Video** Finding one, step by step · Finding inverse functions: linear

► **Video** Checking it · Verifying inverse functions by composition

≡ **Article** The method, written out · Intro to inverse functions

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✍ **Exercise** Finding one · Find inverses of linear functions

✍ **Exercise** Checking by composition · Verify inverse functions

- 1 **Check the function is one-to-one.** If it is not, there is nothing to find yet.
- 2 **Write $y = f(x)$.**
- 3 **Solve for x in terms of y .** This is the only step that is real work.
- 4 **Interchange x and y .** The input goes back to being called x .
- 5 **Write the result as the inverse.**

$$y = f^{-1}(x)$$

- 6 **Check by composing both ways.** You should get x back each time. This catches almost every slip.

$$F(X) = 3X + 4$$

$$y = 3x + 4$$

Start from the definition.

$$x = \frac{y - 4}{3}$$

Solve for x.

$$f^{-1}(x) = \frac{x - 4}{3}$$

Swap the names.

$$f^{-1}(f(x)) = \frac{(3x + 4) - 4}{3} = x$$

The check. It returns x, so it is right.

WHEN THERE IS NO INVERSE YET

Restrict the domain until the function is one-to-one. $f(x) = (x - 1)^2$ has no inverse over all the reals, because it repeats every output. Keep only $x \geq 1$ and it has one.

THE LINEAR CASE, WORTH KNOWING BY SIGHT

$$f(x) = ax + b \quad \Rightarrow \quad f^{-1}(x) = \frac{x - b}{a}$$

WHAT AN INVERSE IS FOR

To solve $f(x) = k$, apply the inverse to both sides. That is the job it does.

WATCH OUT

- Swapping the names before solving. Solve first, rename second.
- Skipping the check. It takes one line and catches a sign error every time.
- Restricting the domain and then forgetting to say so in the answer.

2.6 The rules of logarithms

LEARN IT ON KHAN ACADEMY

- ▶ **Video** The logarithm rules · Intro to logarithm properties (1 of 2)
 - ▶ **Video** The rule that solves equations · Using the logarithmic power rule
 - ▶ **Video** The change of base formula for logarithms
 - ≡ **Article** The rules, written out · Intro to logarithm properties
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PRACTISE IT ON KHAN ACADEMY

- ✍ **Exercise** Expanding and condensing · Use the properties of logarithms
 - ✍ **Exercise** Evaluate logarithms: change of base rule
-

THE THREE RULES

$$\log_b(MN) = \log_b M + \log_b N$$

$$\log_b \frac{M}{N} = \log_b M - \log_b N$$

$$\log_b(M^n) = n \log_b M$$

THE ONE THAT DOES NOT EXIST

There is no rule for $\log(M + N)$. A sum inside a logarithm does not come apart. Nothing you do to it will shorten it.

THE ONES THAT FOLLOW FROM THE DEFINITION

$$\log_b 1 = 0 \quad \log_b b = 1$$

$$\log_b(b^x) = x \quad b^{\log_b x} = x$$

- 1 **To expand, factor the argument first.** A product becomes a sum, a quotient a difference.

- 2 **Then bring any exponent down as a coefficient.** That is the power rule doing the work.
- 3 **To condense, run the three rules backwards.** Power rule first, then the product rule, then the quotient rule. In that order.
- 4 **The power rule is the one that solves equations.** It brings the unknown down out of the exponent, where you can reach it.
- 5 **Change the base when the calculator has not got it.**

$$\log_b M = \frac{\ln M}{\ln b}$$

- 6 **Check the argument is positive before applying any of it.** A logarithm of zero or of a negative number does not exist.

EXPAND $\log\left(\frac{x^2y}{z}\right)$

$$= \log(x^2y) - \log z$$

Quotient rule first.

$$= \log(x^2) + \log y - \log z$$

Then the product rule on the top.

$$= 2 \log x + \log y - \log z$$

Then the power rule on the first term.

SOLVE $3^x = 50$

$$\ln(3^x) = \ln 50$$

Take the log of both sides.

$$x \ln 3 = \ln 50$$

The power rule brings x down.

$$x = \frac{\ln 50}{\ln 3} \approx 3.56$$

Divide, then use the calculator once.

WATCH OUT

- Inventing a rule for $\log(M + N)$. There is not one.
- $\log(M) \log(N)$ is not $\log(MN)$. The product rule turns a product into a *sum*.
- Applying a rule before checking the argument is positive.

2.7 Test yourself

Do this week's homework on the course platform, then check it against the answers when they are posted. That is the test. Everything above is how to be ready for it.



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